

Sanjay Acharjee

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Physical AI | Field Robotics | 3D Perception | Active Perception | Mobile Manipulation

Summary

Ph.D. researcher developing robotic perception and autonomy for complex, changing, and unstructured environments. Research spans active perception, mobile manipulation, 3D scene understanding, visual servoing, robot simulation, reality capture, computer vision, and generative AI. Current work uses **Husky A300 + UR5e** and **LeRobot SO-101** platforms for embodied perception, navigation, and robot control.

Education

The University of Texas at Arlington

Arlington, TX

Ph.D. in Civil Engineering, CGPA: 4.00/4.00

Jan. 2024 – Expected Fall 2027

- **Graduate Certificate:** Artificial Intelligence & Deep Learning, May 2026.
- **Relevant Coursework:** Robotics, Neural Networks, Computer Vision, Machine Learning, Artificial Intelligence, Data Analysis & Modeling, Advanced Building Information Modeling.

Bangladesh University of Engineering and Technology (BUET)

Dhaka, Bangladesh

B.Sc. in Civil Engineering, CGPA: 3.44/4.00

May 2023

Technical Skills

Programming: Python, C++, MATLAB, Git, Linux/Ubuntu

Robotics: ROS 2, Nav2, TF2, URDF/Xacro, MoveIt, forward/inverse/differential kinematics, Jacobian control, visual servoing, path planning, A*/Dijkstra

Simulation & Robot Learning: Gazebo, MuJoCo, Hugging Face LeRobot, Sim2Real workflows

AI / Computer Vision: PyTorch, TensorFlow, OpenCV, Hugging Face, transformers, vision-language models, diffusion models, pose estimation

3D Perception: COLMAP, Open3D, point clouds, SfM/MVS, RGB-D sensing, LiDAR, camera geometry, 3D reconstruction

Digital Construction: BIM/VDC, Scan-to-BIM, Autodesk Revit, Dynamo, IfcOpenShell, Trimble RealWorks

Hardware: Clearpath Husky A300, UR5e, LeRobot SO-101, RGB/RGB-D cameras, Trimble X9, Parrot Anafi USA, Autel Evo Max 4T

Research Experience

University of Texas at Arlington – iSET Lab

Arlington, TX

Graduate Research Assistant

Jan. 2024 – Present

- **Active Perception / Mobile Manipulation:** Developed *DeltaSeek*, an embodiment-aware sensing framework using a **Husky A300 + UR5e**. Evaluated **240 navigable base poses** (60 positions $\times$ 4 yaw orientations) across five articulated arm postures; wrist-mounted sensing observed changes inaccessible from chassis viewpoints and reduced median travel from **15.2 m to 6.0 m** for mutually observable changes. **Paper | Code**
- **Physical AI / Field Robotics Platform:** Developing a multi-camera **Husky A300 + UR5e** research platform using ROS 2, Nav2, TF2, Gazebo, and MoveIt for closed-loop active perception, autonomous navigation, 3D sensing, and mobile manipulation in changing environments.
- **3D Multimodal Perception:** Developed an SfM/MVS pipeline fusing RGB and thermal drone imagery to reconstruct dense 3D thermal profiles for spatially localized building inspection and anomaly detection.
- **Generative / Multimodal AI:** Developed scene-graph-guided pipelines for extracting industrial hazards from historical accident narratives and generating/evaluating visual safety scenarios using LLMs, vision-language models, and diffusion models.

AIEdgeInside

Remote, Tokyo, Japan

Machine Learning Engineer – Full-time

Jun. 2023 – Dec. 2023

- Developed computer-vision/OCR pipelines for handwritten table documents using transformer-based models including TrOCR, automating cell-level information extraction.
- Developed generative-image pipelines for virtual try-on applications and realistic apparel synthesis.

Selected Robotics Project

LeRobot SO-101 Kinematics & Visual Servoing | *MuJoCo, LeRobot, OpenCV, NumPy, SymPy*

2025–2026

- Built a robotics control stack with analytical forward kinematics, Jacobian-based differential inverse kinematics, singularity monitoring, predictive joint-limit safety, and closed-loop image-based visual servoing.
- Validated algorithms in **MuJoCo** before deploying the same control framework to the physical **LeRobot SO-101**; implemented perception-to-control pipelines that convert image-space target errors into joint commands for autonomous end-effector positioning. **GitHub**

Digital Construction & Reality Capture

University of Texas at Arlington

Graduate Teaching Assistant - Advanced BIM; Drones & Advanced Construction Technology

Arlington, TX

2024 – 2025

- Delivered technical instruction in **Dynamo-based BIM automation** and **Reality Capture / Scan-to-BIM**; led hands-on **Trimble X9 LiDAR** workflows covering point-cloud acquisition, Trimble RealWorks processing, and integration into Autodesk Revit.
- Trained students in safe drone operation and field data collection for construction applications, including **FAA Part 107 compliance**, pre-flight planning, airspace/safety considerations, and supervised flight exercises.

Selected Publications

- **Acharjee, S., Sakib, M. N. DeltaSeek: Toward Active Perception in Evolving Construction Environments.** arXiv:2609.33836; IROS Workshop 2026. **Paper | Code**
- **Acharjee, S., Khan Ratul, A., Patiño, D., Sakib, M. N. Floorplan Reconstruction through Field Prediction and Geometry-Aware Vectorization.** Submitted to AAAI 2027.
- **Khan Ratul, A., Acharjee, S., Park, S., Sakib, M. N. Multi-agent human-AI collaborative framework for converting hand-drawn floor plans to 3D BIM.** *Automation in Construction*, 2026.
- **Acharjee, S., Khan Ratul, A., Patiño, D., Sakib, M. N. Scene Graph-Guided Generative AI Framework for Synthesizing and Evaluating Industrial Hazard Scenarios.** arXiv:2511.13970, 2025.
- **Acharjee, S., Khan Ratul, A., Aryal, A., Sakib, M. N. SfM Meets Thermal: Reconstructing 3D Thermal Profiles of Buildings Using Multimodal Drone Data.** ASCE International Conference on Computing in Civil Engineering, 2025.
- **Acharjee, S., Khan Ratul, A., Patiño, D., Sakib, M. N. Data-Driven Safety: Leveraging Historical Accident Data and Generative AI for Generating Workplace Hazard Visualizations.** ASCE International Conference on Computing in Civil Engineering, 2025.
- **Nahin, S. K., Acharjee, S., Saha, S., Das, A., Hossain, S., Haque, M. A. Human sleeping pose estimation from IR images for in-bed patient monitoring using image processing and deep learning techniques.** *Heliyon*, 2024.

Additional Research Project

3D Thermal Profiling from Multimodal Drone Data | COLMAP, Open3D, SfM/MVS, RGB + LWIR

2024–2025

- Reconstructed building-scale 3D geometry from drone imagery and mapped thermal information into dense point clouds for spatially localized inspection and energy diagnostics.

Honors & Certifications

- **Champion**, IEEE Video and Image Processing Cup (VIP Cup), 2021.
- **Second Runner-up**, IEEE Video and Image Processing Cup, 2022.
- **First Runner-up**, DL Sprint Bengali Automatic Speech Recognition Competition, 2022.
- **Graduate Dean's STEM Research Assistantship**, University of Texas at Arlington, 2024 and 2026.
- **FAA Remote Pilot Certificate (sUAS).**